

Reflex Lab 01

Computer Setup Guide

This guide gets your laptop ready before the physical build. The circuit works the same on **Windows** and **Mac** — the differences are mostly USB ports, browser choice, and how you launch the local dashboard.

Quick Start

Already know Arduino IDE? Use this checklist:

Step	Windows	Mac
Browser	Chrome or Edge	Chrome or Edge
Arduino IDE	Install Arduino IDE 2.x	Install Arduino IDE 2.x
ESP32 board pkg	esp32 by Espressif Systems	esp32 by Espressif Systems
Port format	COM3, COM6, etc.	/dev/cu.usbserial-* etc.
Firmware path	kernels\threshold_reflex_v0\...\threshold_reflex_v0.ino	kernels/threshold_reflex_v0/...\threshold_reflex_v0.ino
Serial speed	115200 baud	115200 baud
Dashboard	start-reflex-dashboard.bat	python3 dashboard.py

Mac note: The Mac bench path is marked *not yet Monogate bench-verified* until a tester records a full Mac upload and dashboard run.

Preferred Browser

Use **Chrome** or **Edge** throughout this course.

- Safari is fine for reading the lab packet, but **do not use Safari for Web Serial workflows**.
- Web Serial is a Chromium-browser feature — Chrome and Edge are the supported student path.
- The local Python dashboard does the actual serial reading in this lab, but Chrome/Edge keeps things consistent with future browser-serial labs.

Install Arduino IDE

1. Download Arduino IDE 2.x from:
2. Install it normally.
3. Open Arduino IDE **once** before plugging in the ESP32.

```
https://www.arduino.cc/en/software
```

Add ESP32 Board Support

1. Open Arduino IDE.
2. Go to **File > Preferences**.
3. In *Additional boards manager URLs*, add the URL below.
4. Go to **Tools > Board > Boards Manager**.
5. Search for **esp32**.
6. Install **esp32 by Espressif Systems**.

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```

Open the Firmware

Open this file in Arduino IDE:

```
kernels/threshold_reflex_v0/esp32/threshold_reflex_v0/threshold_reflex_v0.ino
```

Then select the board:

```
Tools > Board > esp32 > ESP32 Dev Module
```

Tip: If your board has a more specific name use the vendor's name. For the generic ESP32 DevKit, *ESP32 Dev Module* is the right starting choice.

Find Your ESP32 Port

Plug in the ESP32 with a **data-capable USB cable**, then go to **Tools > Port** in Arduino IDE.

Windows	Mac
COM3	/dev/cu.usbserial-0001
COM6	/dev/cu.SLAB_USBtoUART
COM7	/dev/cu.wchusbserial*

Mac tip: When both a *cu* and *tty* port appear, always choose the **cu** port.

Upload Firmware

1. Confirm the board is safely wired (or disconnect extra parts).
2. Select the **ESP32 board** and **serial port**.
3. Click **Verify**.
4. Click **Upload**.
5. If upload fails while connecting, hold the **BOOT** button when Arduino says "Connecting...", then release when writing begins.

After upload, open Serial Monitor at **115200 baud**. You should see JSONL-style frames streaming in.

Run the Local Dashboard

The dashboard displays ESP32 serial frames and lets you export evidence. **Close Arduino Serial Monitor before starting** — only one program can read the serial port at a time.

Windows

Double-click from the course folder:

```
courses\001-guarded-reflex-loops\lab-bundle\start-reflex-dashboard.bat
```

If your ESP32 is not on COM6, set the port from PowerShell:

```
$env:MG_SERIAL_PORT = "COM7"
.\courses\001-guarded-reflex-loops\lab-bundle\start-reflex-dashboard.ps1
```

Mac

The one-click launcher is Windows-only for this release. On Mac, run the dashboard from Terminal:

```
python3 -m pip install pyserial python3 electronics/dashboards/esp32-arduino/dashboard.py \
--port /dev/cu.usbserial-0001 --baud 115200
```

Replace `/dev/cu.usbserial-0001` with your actual ESP32 port.

Then open Chrome or Edge to:

```
http://127.0.0.1:5191/visualizer
```

Troubleshooting

● ESP32 powers on but no port appears

- Try another USB cable — many cables are charge-only.
- Try another USB port on the laptop.
- Unplug the ESP32, reopen Tools > Port, plug back in, and watch what appears.
- If your board uses a CP210x or CH340 chip, install the driver from the chip vendor.
- **Mac only:** If a driver install asks for approval, check System Settings > Privacy & Security, approve, and restart if prompted.

● Upload fails

- Confirm *ESP32 Dev Module* (or correct variant) is selected.
- Confirm the correct port is selected.
- Close Serial Monitor and the dashboard before uploading.
- Hold **BOOT** during "Connecting..." if your board needs manual boot mode.
- Try a lower upload speed in Arduino IDE if the board is unreliable.

● Dashboard says Offline

- Close Arduino Serial Monitor.
- Confirm firmware was uploaded successfully.
- Confirm the dashboard port matches the Arduino port.
- Confirm baud rate is 115200.
- Install pyserial if the dashboard reports it is missing: `python3 -m pip install pyserial`

● Browser doesn't see serial features

- Use Chrome or Edge — do not use Safari for Web Serial.
- If using the local Python dashboard, the browser only shows the page; Python does the serial reading.

● Buzzer or LED doesn't respond

- Recheck the simulator wiring checklist before blaming the computer.
- Confirm ESP32 GND, pot ground, LED cathode, button ground, and buzzer ground all share the same ground net.
- Confirm the LED has its 220 or 330 ohm resistor.
- Confirm the buzzer path uses the 1 kOhm resistor.
- Confirm the button goes to GPIO27 and ground (firmware uses it to mute only the buzzer branch).

What to Record

Capture the following for bench validation (Mac and Windows):

- ✓ Screenshot of Arduino IDE **board selection**
- ✓ Screenshot of Arduino IDE **port selection**
- ✓ Successful **upload message**

- ✓ Dashboard showing **live frames**
- ✓ Exported **JSONL trace**
- ✓ Note with: board model, USB cable type, browser, and OS version